AP CALCULUS AB

Unit 1 • Lesson 1.13 practice • Answer key

Lesson 1.13 • Removing Discontinuities

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Let $g(x) = \dfrac{x^2 - 16}{x - 4}$. What value of $g(4)$ makes g continuous at $x = 4$?

(A) 0
(B) 4
(C) 16
(D) 8

Solution

Answer: (D). $\lim_{x \to 4}(x + 4) = 8$.

Question 2

NO CALCULATOR

For what value of k is $f(x) = \begin{cases} kx^2, & x \leq 1 \\ x + 3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

Solution

Answer: (A). $k(1)^2 = 1 + 3$, so $k = 4$.

Question 3

NO CALCULATOR

For what value of a is $f(x) = \begin{cases} ax + 2, & x < 3 \\ x^2 - a, & x \geq 3 \end{cases}$ continuous?

(A) $\dfrac{11}{4}$
(B) $\dfrac{7}{4}$
(C) $\dfrac{7}{2}$
(D) 7

Solution
Answer: (B). $3a + 2 = 9 - a$, so $4a = 7$ and $a = \frac{7}{4}$.

Question 4

NO CALCULATOR

Let $f(x) = \dfrac{\sin x}{x}$ for $x \neq 0$. What value of $f(0)$ makes f continuous at 0?

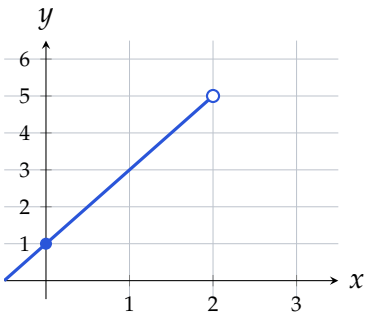
(A) 0
(B) undefined
(C) 1
(D) π

Solution
Answer: (C). $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

Question 5 (graph)

NO CALCULATOR

For $x < 2$, the graph of f is the line shown, through $(0, 1)$ and $(2, 5)$. For $x \geq 2$, $f(x) = x^2 + k$. For what value of k is f continuous at $x = 2$?



(A) 5
(B) -1
(C) 9
(D) 1

Solution
Answer: (D). From the graph, $\lim_{x \to 2^-} f(x) = 5$. Continuity needs $f(2) = 4 + k = 5$, so $k = 1$.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \begin{cases} 2x + a, & x < 1 \\ bx^2 + 1, & x \geq 1 \end{cases}$ where a and b are constants, and $f(2) = 9$.

(a) Find b.
(b) Find the value of a that makes f continuous at $x = 1$.
(c) With these values, verify that f is continuous at $x = 1$ using the definition.

Solution
(a) $f(2) = 4b + 1 = 9$, so $b = 2$.
(b) $\lim_{x \to 1^-} f(x) = 2 + a$ and $\lim_{x \to 1^+} f(x) = b + 1 = 3$. Then $2 + a = 3$, so $a = 1$.
(c) $f(1) = 2(1)^2 + 1 = 3$; $\lim_{x \to 1^-} f(x) = 3$ and $\lim_{x \to 1^+} f(x) = 3$, so $\lim_{x \to 1} f(x) = 3 = f(1)$: f is continuous at $x = 1$.